KEVIN KWAN

Software Engineer II | Conversational AI & Voice Automation
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SUMMARY

Software Engineer II specializing in conversational AI and voice automation, building and operating production AI voice agents and IVR platforms on AWS that resolve real customer calls at scale. Georgia Tech Computer Science, *summa cum laude*. Track record of self-service impact, migration delivery, and reliability leadership in high-volume production environments.

EDUCATION

Georgia Institute of Technology — Atlanta, GA August 2021 – May 2024

- Bachelor of Science in Computer Science; *summa cum laude* GPA: 3.9
- Zell Miller Scholarship, Graduated with Highest Honors, Dean's List, Faculty Honors

TECHNICAL SKILLS AND KNOWLEDGE

Languages: Java, Python, C#, JavaScript, TypeScript, SQL, Lua, C
Conversational AI & Voice: Sierra AI, Amazon Connect, Amazon Lex, Cyara
Cloud & Backend: AWS (Lambda, Connect, Lex, DynamoDB), Azure, Google Cloud, Serverless Architectures, REST APIs
Security: AES 256 Encryption, Secure API Design, PII Protection
DevOps & Tooling: Git, Terraform, CI/CD, GitHub, Azure DevOps, Docker, Node.js, SonarQube, Cursor, Claude Code
Databases: DynamoDB, MongoDB, MySQL
Frontend: React, Next.js, HTML, CSS
Practices: Agile/Scrum, SDLC, Application Resiliency, Production Support

WORK EXPERIENCE

Software Engineer II — GEICO July 2024 – Present
Atlanta, GA (Remote) | Promoted from Software Engineer I

- Co-built and operate the Sierra AI billing voice agent (live since May 2026), handling ∼30,000 GEICO billing calls per week — payments, rescheduling, AutoPay, billing questions — in English and Spanish; contained ∼51% of 436,112 conversations without a live-agent transfer (May–Oct 2026), trending 48% → 52% as weekly volume grew ∼10×.
- Delivered end-to-end make-payment for the agent (saved method, pay-now or scheduled, debit authorization, declined-/expired-card recovery, payments on policies pending cancellation): 22,600+ payments at 92–95% authorization success; shipped net-new Pause AutoPay, 82,800 reschedules, and 192,000 info answers at 98% tool success.
- Re-architected the billing agent from regex-/classifier-based routing to fact-based journeys (178 files, ∼20,000 net lines removed) and built simulation-based regression quality gates (5-of-5 pass standard) plus automated code-review follow-up, keeping high-velocity releases safe.
- Fixed 86 of 91 production and UT1 billing bugs at the root cause (median 3 days from report to fix); authored 468 PRs — 309 of 559 commits to the billing agent since May, its largest contributor — and 294 production builds across 45 repos; reviewed 653 PRs for 12+ engineers and onboarded two engineers onto the Sierra platform.
- Key contributor to migrating GEICO's enterprise IVR platform from Cisco UCCE and Microsoft Nuance to Amazon Connect on AWS; developed Java traffic-redirection logic to shift a controlled percentage of live caller traffic gradually, and delivered legacy endpoint migration plus a Lex pattern cutover on schedule.
- Improved the AWS IVR billing and payments experience (Dec 2025–Apr 2026): built the zero-amount-due flow, reworked billing error handling so declined cards and banks retry instead of transferring to an agent, and delivered new card tokenization.
- Delivered customer self-service payment workflows on the AWS IVR at ∼60% containment across very high call volume, and led the Proof of Coverage IVR solution at ∼80% containment; own its disaster-recovery rebuild pipeline.
- Designed and implemented AWS Lambda functions and Amazon Lex bots (JavaScript code hooks) connecting internal APIs with Amazon Connect contact flows; built an asynchronous, high-throughput Lambda orchestration system backed by DynamoDB queues processing tens of thousands of concurrent requests.
- Implemented secure internal APIs exposing sensitive customer data using AES encryption to protect PII and payment information (credit card and bank data); maintained Terraform infrastructure pipelines for secure, repeatable deployments.
- Extensive on-call and production support: release validation, live IVR testing, outage response and rollbacks, post-mortems, and a 41,829-call conversation-experience review that ranked and closed the agent's biggest customer-experience gaps.

Software Engineer Intern — Publicis Sapient (Publicis Groupe) June 2023 – August 2023
Atlanta, GA

- Engineered an AI-powered conversational chatbot with memory and agent tools using Google Vertex AI's LLM, LangChain, MongoDB, and APIs for the Travel & Hospitality industry, giving tourists accurate, up-to-date answers and recommendations for Atlanta attractions, hotels, and restaurants.
- Architected and deployed RESTful APIs using Flask and Python connecting the frontend to backend MongoDB databases and the AI chatbot endpoint to generate and retrieve responses.
- Improved the accuracy of the bot's responses up to 91% through rigorous testing and validation.